

MACHINE LEARNING FOR UIC STUDENT PREDICTION

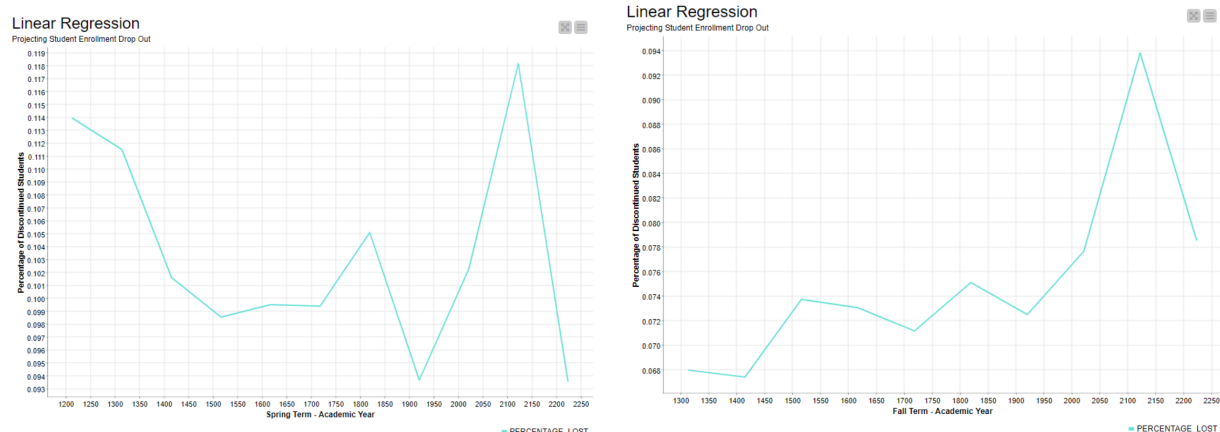
TREE ENSEMBLE WITH LINEAR REGRESSION PROJECTION TO ADJUST THE THRESHOLD FOR THE SENSITIVITY OF CLASSIFICATION

INTRODUCTION

Within the KNIME workflow, there are two different versions of the implementation of the Tree Ensemble model. Prior to the most recent data table load on 6/21/2023, which took place shortly after the census snapshot on 6/18/2023, the testing, training, and comparison of various models were conducted using data that lacked the records of summer students. As a result, before the census snapshot on 6/18/2023, these students were not included in the data table, causing significant and abrupt decreases in classification numbers that deviated from the linear trend observed in historical data.

Additionally, a consistent bias was observed in the classifications, characterized by a constant difference between the actual number of discontinued students and the predicted values. To address this issue and account for the linear trend in the data, I employed a linear regression technique to create a projection estimating the number of expected discontinued students based on historical data.

In summary, the inclusion of the missing summer student data and the utilization of linear regression allowed for a more accurate capture of the linear trend within the dataset, mitigating the drop-offs in classification numbers and reducing the bias observed in the predictions.



This is the actual percentage of students that discontinued for all academic years including the projected percentage of students to drop in the academic year 2223.

I	ACAD...	S	SEM...	D	TOTAL_NUM_STUDENTS_DISCONTINUED	I	TOTAL_NUM...	D	PERCENTAGE...	D	PREDICTED...	D	ADJUSTMENT...
1213		1			1,725.24		15684		0.11		0.042		2.644
1314		1			1,722.38		15658		0.11		0.042		2.638
1415		1			1,744.05		15855		0.11		0.041		2.657
1516		1			1,791.57		16287		0.11		0.041		2.67
1617		1			1,654.1		16541		0.1		0.041		2.469
1718		1			1,816		18160		0.1		0.041		2.469
1819		1			1,924		19240		0.1		0.041		2.445
1920		1			2,005.7		20057		0.1		0.042		2.392
2021		1			2,010.5		20105		0.1		0.042		2.364
2122		1			2,012.5		20125		0.1		0.044		2.262
2223		1			1,997.6		19976		0.1		0.045		2.237
1314		8			1,166.97		16671		0.07		0.02		3.448
1415		8			1,170.26		16718		0.07		0.02		3.483
1516		8			1,230.25		17575		0.07		0.021		3.398
1617		8			1,257.2		17960		0.07		0.021		3.333
1718		8			1,361.36		19448		0.07		0.021		3.271
1819		8			1,454.81		20783		0.07		0.021		3.333
1920		8			1,514.94		21642		0.07		0.021		3.398
2021		8			1,534.47		21921		0.07		0.021		3.382
2122		8			1,782.32		22279		0.08		0.023		3.448
2223		8			1,744.56		21807		0.08		0.024		3.333

This table shows the Academic year, semester code, total number of students that actually discontinued, the the total number of students enrolled for that academic year, the moving average of the actual percentage of students that discontinued, the moving average of predicted students to discontinue, and the moving average of the adjustment ration that should be applied to probability threshold for each year.

The presented analysis showcases a linear projection illustrating the anticipated percentage of student dropouts throughout the spring and fall terms of the academic year 2223. To enhance the generalizability of these projections and effectively capture the linear trend, a moving average approach has been employed over the actual percentage of students lost. Then by calculating the ratio between the observed percentage of students lost and the corresponding predicted percentage of students lost as generated from the tree ensemble model, an adjustment ratio has been derived which can be seen in the column called “ADJUSTMENT_RATIO”. This adjustment ratio holds potential for modifying the sensitivity threshold in classification, leveraging the probability distribution generated by the random forest model. As such, this methodology offers the opportunity to refine the classification process by incorporating insights obtained from the linear projection and adjustment ratio analysis.

D	P (SUCCESSFUL_OUTCOME_TARGET_BY_TERM=0)_probability	S	Prediction (SUCCESSFUL_OUTCOME_TARGET_BY_TERM)	D	Prediction (SUCCESSFUL_OUTCOME_TARGET_BY_TERM) (Confidence)
0		1		1	
0.729		0		0.729	
0.63		0		0.63	
0.747		0		0.747	
0.649		0		0.649	
0.748		0		0.748	
0.589		0		0.589	

Seen above is a sample of the column of values for probability measure of classification, the assigned target based on the model, and the confidence of that measure.

The current classification threshold for identifying discontinued students is set at any probability greater than 0.50. However, empirical evidence suggests that this threshold is too high, resulting in an inadequate number of correctly classified students. Therefore, it becomes necessary to lower the threshold to encompass an appropriate number of students.

To adjust the threshold based on the probability distribution, the following steps can be followed:

1. Calculate the mean (mean_prob) and standard deviation (std_prob) of the probability measure derived from the current model.
2. Determine the desired number of predictions that should be included.
3. Utilize the inverse cumulative distribution function (also known as the percent point function) of a normal distribution to identify the corresponding threshold value required to include the desired number of predictions. This function provides the value at which a given probability is achieved in a normal distribution.
4. Calculate the desired probability of inclusion by dividing the number of desired predictions by the total population size (e.g., the number of samples in the dataset).
5. Use the formula $\text{threshold} = \text{mean_prob} + (Z * \text{std_prob})$, where Z represents the inverse of the cumulative distribution function with the desired probability as the argument. This calculation yields the threshold value relative to the probability measure's distribution.

By employing this approach to adjust the threshold, the number of predictions can be increased or decreased based on the desired quantity and the characteristics of the probability measure's distribution. It is important to note that this method assumes the probability measure follows a normal distribution or exhibits similar properties.

This technique notably enhanced the model's performance, enabling more accurate classification of the appropriate number of students for discontinuation. However, further examination revealed that the probability distribution deviates from the assumed normality.

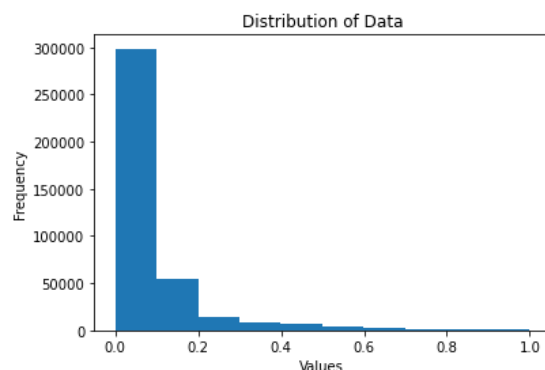
NEXT STEPS SUGGESTIONS

```
df = pd.read_parquet("distribution.parquet")

import matplotlib.pyplot as plt

# Assuming your dataframe is named 'df' and the column you want to visualize is
column_data = df['P (SUCCESSFUL_OUTCOME_TARGET_BY_TERM=0)_probability']

# Create a histogram using Matplotlib
plt.hist(column_data, bins=10) # Adjust the number of bins as needed
plt.xlabel('Values')
plt.ylabel('Frequency')
plt.title('Distribution of Data')
plt.show()
```



Probability of Discontinuation Distribution

The analysis reveals that the probability distribution exhibits a left-skewness and is non-normal. Consequently, adjusting the threshold solely based on the mean and lowering it uniformly may result in uneven classifications. This implies that, depending on the specific data subset distributions, the classification may still suffer from under-prediction or over-prediction issues. Although the aforementioned method served as an initial attempt, it is advisable to explore alternative approaches to improve classification performance and threshold tuning. If the intention is to continue using the ratio between the actual and predicted percentage of student loss to adjust the threshold, a more advanced approach is recommended to achieve a more accurate representation of the probability distribution. One potential avenue worth exploring involves utilizing the Probability Density Function (PDF).

To determine the proportion of samples falling within a specified range of values using the PDF, integration of the PDF over that range is required. Here is a step-by-step guide:

1. Define the desired range of values for calculating the proportion (e.g., from a to b).
2. Employ the PDF formula of the probability distribution to obtain the density function corresponding to the distribution in question.
3. Integrate the density function over the range from a to b, as the integral represents the area under the curve between those limits.
4. The resulting integral value represents the proportion (or probability) of samples falling within the specified range.
- 5.

The exact integration process will depend on the specific probability distribution being utilized. Let's consider an example using the normal distribution:

Assuming a normal distribution with mean μ and standard deviation σ , the PDF is given by:

$$f(x) = (1 / (\sigma * \sqrt{2\pi})) * e^{-(x - \mu)^2 / (2\sigma^2)}$$

To determine the proportion of samples falling between two values, denoted as a and b, you would integrate the PDF over that range:

$$P(a \leq x \leq b) = \int[a \text{ to } b] f(x) dx$$

This integral will yield the proportion of samples falling within the range [a, b] according to the normal distribution.

By iteratively integrating over the PDF using the mean (μ) as the starting point and setting b as (standard deviation * adjustment ratio), adjustments can be made until the obtained sample size aligns with the predicted enrollment number. This stepwise approach helps approximate the appropriate threshold by considering the underlying PDF characteristics.

OUTPUT

UNDERGRADUATE CLASSIFICATION TO DISCONTINUE

Scorer View

Confusion Matrix

Rows Number : 105813	0 (Predicted)	1 (Predicted)	
0 (Actual)	985	8634	10.24%
1 (Actual)	399	95795	99.59%
	71.17%	91.73%	

Class Statistics

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
0	985	399	95795	8634	10.24%	71.17%	10.24%	99.59%	17.90%
1	95795	8634	985	399	99.59%	91.73%	99.59%	10.24%	95.50%

Overall Statistics

Overall Accuracy	Overall Error	Cohen's kappa (κ)	Correctly Classified	Incorrectly Classified
91.46%	8.54%	0.160	96780	9033

Tree Ensemble Actual Vs Predicted

Undergraduate Classification Discontinued Spring



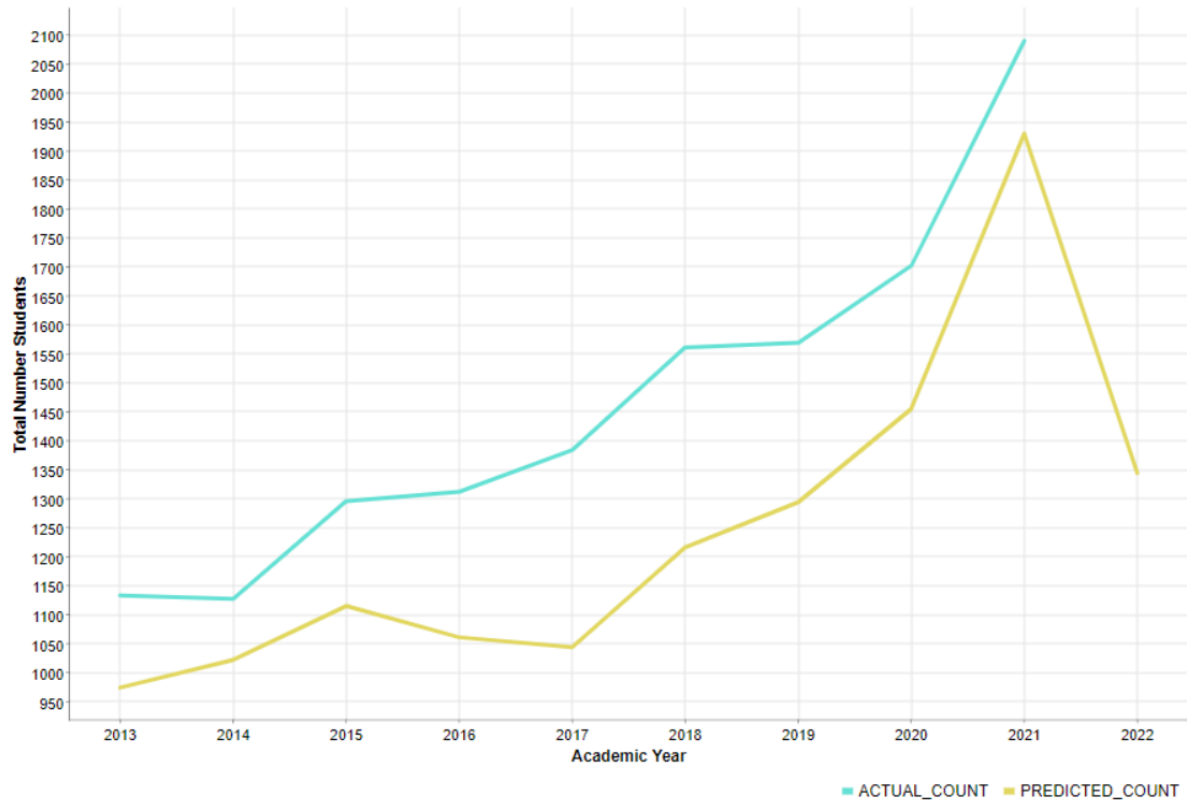
Upon analysis, it becomes evident that the predicted count still exhibits a slight bias towards underprediction. This bias is likely attributable to the non-normal distribution of the probability. Consequently, while the adjustment threshold has improved the bias to some extent, it remains insufficient for achieving accurate predictions. Thus, alternative methods need to be explored to enhance the accuracy of the threshold and address the existing biases effectively.

First(YEAR_CD)	ACAD_YEAR_CODE	SEMESTER_CD	ACTUAL_COUNT	PREDICTED_COUNT
2013	1213	1	1787	1519
2014	1314	1	1746	1457
2015	1415	1	1611	1361
2016	1516	1	1605	1350
2017	1617	1	1646	1265
2018	1718	1	1805	1399
2019	1819	1	2022	1643
2020	1920	1	1879	1369
2021	2021	1	2056	1536
2022	2122	1	2378	1905
2023	2223	1	?	1143

The table of actual vs predicted values. Showing the predicted value for the number of discontinued students for next fall term.

Tree Ensemble Actual Vs Predicted

Undergraduate Classification Discontinued Fall



In this observation, it becomes evident that bias is less prominent among the undergraduate students. This is likely because the probability distribution they encounter may not exhibit a strong left skew, unlike the one experienced during the spring season.

First(YEAR_CD)	ACAD_YEAR_CODE	SEMESTER_CD	ACTUAL_COUNT	PREDICTED_COUNT
2013	1314	8	1133	974
2014	1415	8	1127	1022
2015	1516	8	1296	1115
2016	1617	8	1312	1061
2017	1718	8	1384	1044
2018	1819	8	1561	1216
2019	1920	8	1569	1294
2020	2021	8	1702	1455
2021	2122	8	2090	1930
2022	2223	8	?	1344

The table of actual vs predicted values. Showing the predicted value for the number of discontinued students for next fall term.

GRADUATE CLASSIFICATION TO DISCONTINUE

Scorer View



Confusion Matrix

Rows Number : 44299	0 (Predicted)	1 (Predicted)	
0 (Actual)	1691	3985	29.79%
1 (Actual)	677	37946	98.25%
	71.41%	90.50%	

Class Statistics

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
0	1691	677	37946	3985	29.79%	71.41%	29.79%	98.25%	42.04%
1	37946	3985	1691	677	98.25%	90.50%	98.25%	29.79%	94.21%

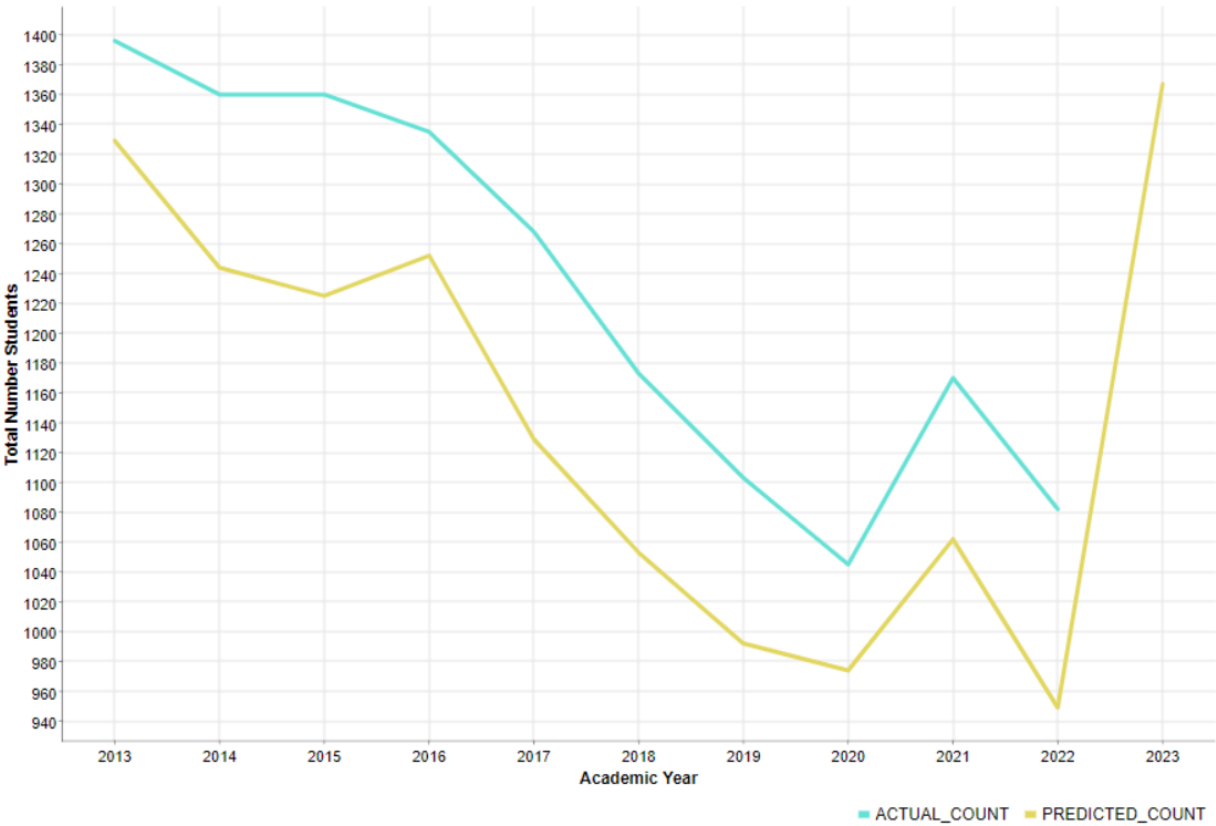
Overall Statistics

Overall Accuracy	Overall Error	Cohen's kappa (κ)	Correctly Classified	Incorrectly Classified
89.48%	10.52%	0.373	39637	4662

Tree Ensemble Actual Vs Predicted



Graduate Classification Discontinued Spring



First(YEAR_CD)	ACAD_YEAR_CODE	SEMESTER_CD	ACTUAL_COUNT	PREDICTED_COUNT
2013	1213	1	1396	1329
2014	1314	1	1360	1244
2015	1415	1	1360	1225
2016	1516	1	1335	1252
2017	1617	1	1268	1129
2018	1718	1	1173	1053
2019	1819	1	1103	992
2020	1920	1	1045	974
2021	2021	1	1170	1062
2022	2122	1	1082	949
2023	2223	1	?	1367

The table of actual vs predicted values. Showing the predicted value for the number of discontinued students for next fall term.

Tree Ensemble Actual Vs Predicted

Graduate Classification Discontinued Fall



First(YEAR_CD)	ACAD_YEAR_CODE	SEMESTER_CD	ACTUAL_COUNT	PREDICTED_COUNT
2013	1314	8	931	895
2014	1415	8	829	799
2015	1516	8	827	823
2016	1617	8	738	688
2017	1718	8	657	615
2018	1819	8	628	596
2019	1920	8	626	635
2020	2021	8	643	596
2021	2122	8	786	707
2022	2223	8	?	800

The table of actual vs predicted values. Showing the predicted value for the number of discontinued students for next fall term.

PROFESSIONAL CLASSIFICATION TO DISCONTINUE

Scorer View

Confusion Matrix

Rows Number : 17339	0 (Predicted)	1 (Predicted)	
0 (Actual)	29	431	6.30%
1 (Actual)	11	16868	99.93%
	72.50%	97.51%	

Class Statistics

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
0	29	11	16868	431	6.30%	72.50%	6.30%	99.93%	11.60%
1	16868	431	29	11	99.93%	97.51%	99.93%	6.30%	98.71%

Overall Statistics

Overall Accuracy	Overall Error	Cohen's kappa (κ)	Correctly Classified	Incorrectly Classified
97.45%	2.55%	0.112	16897	442

Tree Ensemble Actual Vs Predicted

Professional Classification Discontinued Spring



First(YEAR_CD)	ACAD_YEAR_CODE	SEMESTER_CD	ACTUAL_COUNT	PREDICTED_COUNT
2013	1213	1	70	46
2014	1314	1	74	53
2015	1415	1	99	57
2016	1516	1	95	57
2017	1617	1	99	68
2018	1718	1	121	80
2019	1819	1	132	91
2020	1920	1	105	69
2021	2021	1	123	92
2022	2122	1	142	111
2023	2223	1	?	33

The table of actual vs predicted values. Showing the predicted value for the number of discontinued students for next fall term.

Tree Ensemble Actual Vs Predicted

Professional Classification Discontinued Fall



First(YEAR_CD)	ACAD_YEAR_CODE	SEMESTER_CD	ACTUAL_COUNT	PREDICTED_COUNT
2013	1314	8	40	29
2014	1415	8	81	126
2015	1516	8	58	68
2016	1617	8	78	58
2017	1718	8	51	46
2018	1819	8	74	59
2019	1920	8	63	58
2020	2021	8	53	54
2021	2122	8	53	43
2022	2223	8	?	40

The table of actual vs predicted values. Showing the predicted value for the number of discontinued students for next fall term.

LAW CLASSIFICATION TO DISCONTINUE

Scorer View

Confusion Matrix

Rows Number : 1179	0 (Predicted)	1 (Predicted)	
0 (Actual)	3	34	8.11%
1 (Actual)	1	1141	99.91%
	75.00%	97.11%	

Class Statistics

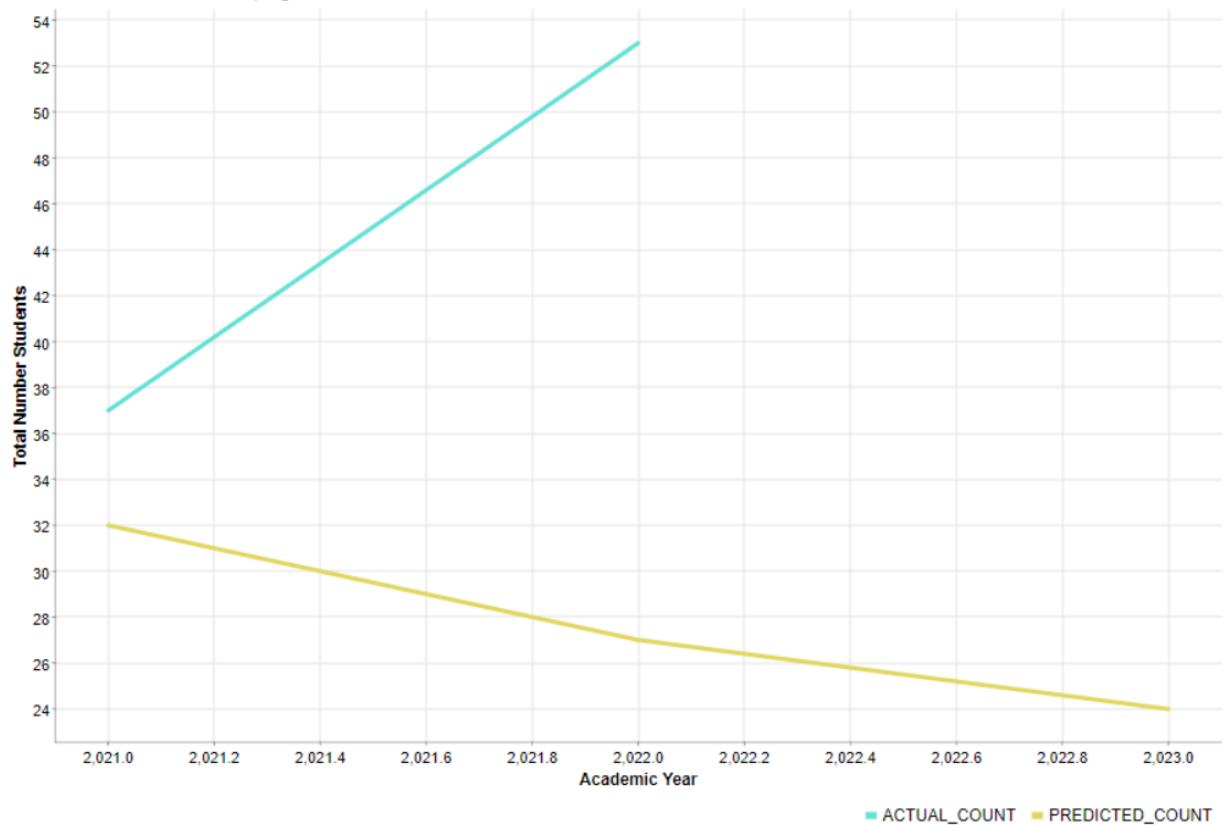
Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
0	3	1	1141	34	8.11%	75.00%	8.11%	99.91%	14.63%
1	1141	34	3	1	99.91%	97.11%	99.91%	8.11%	98.49%

Overall Statistics

Overall Accuracy	Overall Error	Cohen's kappa (κ)	Correctly Classified	Incorrectly Classified
97.03%	2.97%	0.141	1144	35

Tree Ensemble Actual Vs Predicted

Law Classification Discontinued Spring

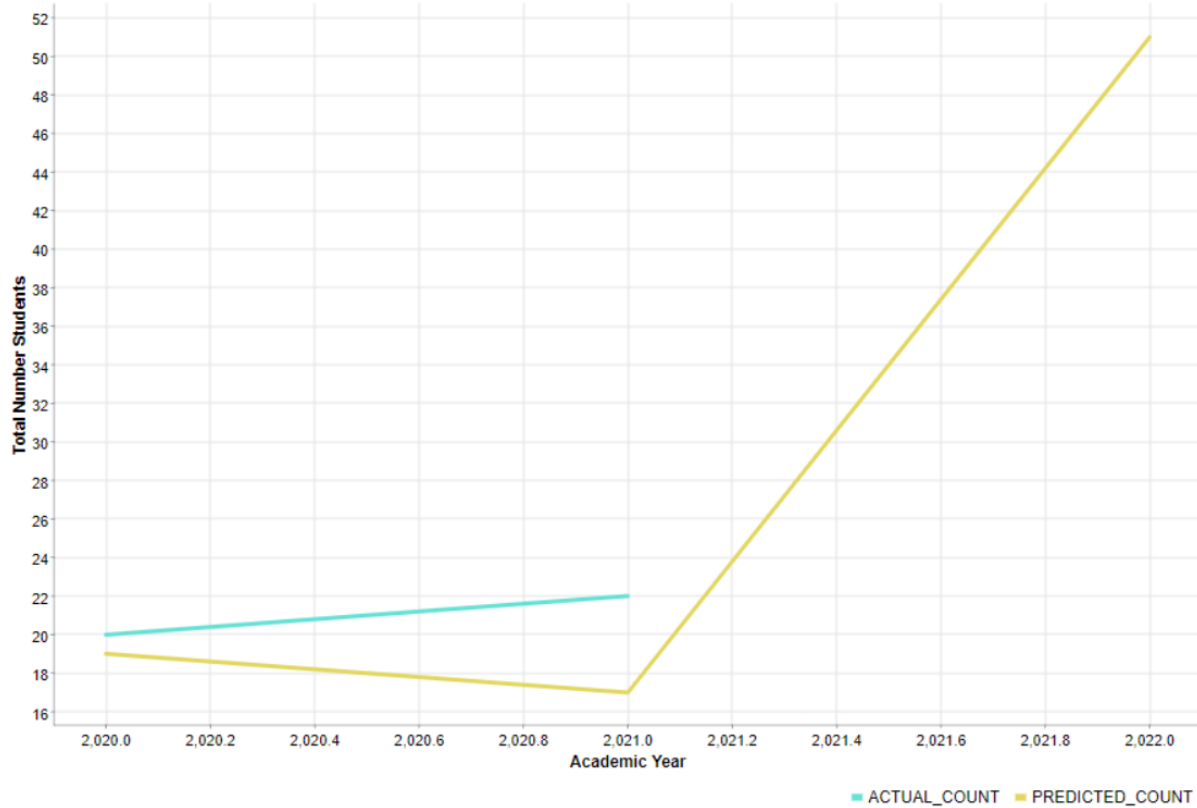


First(YEAR_CD)	ACAD_YEAR_CODE	SEMESTER_CD	ACTUAL_COUNT	PREDICTED_COUNT
2021	2021	1	37	32
2022	2122	1	53	27
2023	2223	1	?	24

The table of actual vs predicted values. Showing the predicted value for the number of discontinued students for next fall term.

Tree Ensemble Actual Vs Predicted

Law Classification Discontinued Fall



First(YEAR_CD)	ACAD_YEAR_CODE	SEMESTER_CD	ACTUAL_COUNT	PREDICTED_COUNT
2020	2021	8	20	19
2021	2122	8	22	17
2022	2223	8	?	51

The table of actual vs predicted values. Showing the predicted value for the number of discontinued students for next fall term.

FINAL RESULTS

TREE_ENSEMBLE_LINEAR_REGRESSION_CLASSIFICATION_TOTAL_BY_TERM.parquet

I ▼ Firs...	I ACAD_...	I SEMESTER_CD	I PREDICTED_COUNT	S STUDENT_TYPE
2023	2223	1	1143	UNDERGRADUATES
2023	2223	1	1367	GRADUATES
2023	2223	1	33	PROFESSIONAL
2023	2223	1	24	LAW
2022	2223	8	1344	UNDERGRADUATES
2022	2223	8	800	GRADUATES
2022	2223	8	40	PROFESSIONAL
2022	2223	8	51	LAW

This is the total classification for the different student levels based on semesters.

I ACAD_...	S STUDENT_TYPE	I Sum(PREDICTED_COUNT)
2223	GRADUATES	2167
2223	LAW	75
2223	PROFESSIONAL	73
2223	UNDERGRADUATES	2487

This is the total predicted number of student to discontinue in the academic year 2223 by student level description.

L ACAD_YR_CD	D TOTAL_STUDENTS	S TYPE
2223	-4,802	DISCONTINUED STUDENTS
2223	-7,807	GRADUATED STUDENTS
2223	35,971	TOTAL STUDENTS

The total number of students can be determined by counting the registration and enrollment figures, while identifying graduate students can be done by checking if the STUDENT_CURR_DEG_IND is "Y," indicating the awarding of a degree. Subtracting these graduate students from the overall enrollment count for the academic year of 2223, along with the predicted number of discontinued students, allows us to calculate the expected total number of students continuing into the next fall term, given the known values.

Row ID	D CONTINUING_STUDENTS
Row0	23,362

Total number of students to continue into the next fall term.

TREE ENSEMBLE WITH DATA AFTER CENSUS SNAPSHOT

In this project, we extensively explored various machine learning models, target variable labeling methods, and aggregation techniques. The key findings of our analysis revealed that

the most effective approach involved aggregating data on a yearly basis, considering the most recent term snapshot for each student in that year, as it resulted in more accurate predictions for the next term. Moreover, adopting a binary target variable that classified next term fall enrollment based on discontinuation yielded the best results. By subtracting the total number of students predicted to discontinue from the known enrollment count for that year and accounting for graduated students, we could determine the number of continuing students. Unlike the previous model, this approach does not employ linear regression to adjust the threshold.

Stratified sampling was employed to balance the dataset, and the attributes used were determined through feature selection (please refer to the feature selection folders for more details). The data was segregated based on student level description, creating subsets for undergraduates, graduates, professional students, law, non-credit, and SCALES. Notably, SCALES represents a discontinued level type, while the enrollment for non-credit was too low to be included. Additionally, the classification reliability for law is limited as we only had three years of data collection. The professional students' trends exhibited volatility, making them unreliable for current classifications. Despite these limitations, the classifications for these subsets are included in the analysis.

OUTPUT

UNDERGRADUATE CLASSIFICATION TO DISCONTINUE

Scorer View

Confusion Matrix



Rows Number : 61784	0 (Predicted)	1 (Predicted)	
0 (Actual)	4373	2976	59.50%
1 (Actual)	815	53620	98.50%
	84.29%	94.74%	

Class Statistics

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
0	4373	815	53620	2976	59.50%	84.29%	59.50%	98.50%	69.76%
1	53620	2976	4373	815	98.50%	94.74%	98.50%	59.50%	96.59%

Overall Statistics

Overall Accuracy	Overall Error	Cohen's kappa (κ)	Correctly Classified	Incorrectly Classified
93.86%	6.14%	0.665	57993	3791

Tree Ensemble Actual Vs Predicted

Undergraduate Classification Discontinued Summer

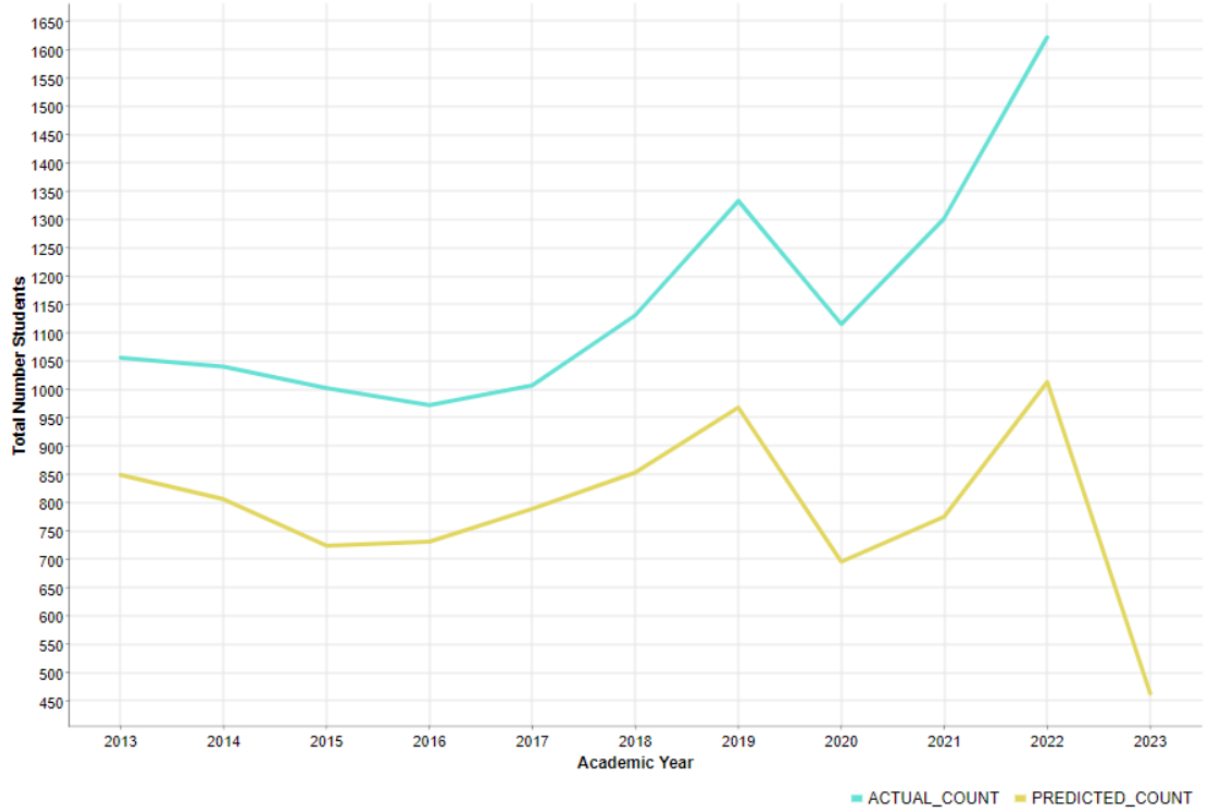


I First(YEAR_CD)	I ACAD_YEAR_CODE	I SEMESTER_CD	I ACTUAL_COUNT	I PREDICTED_COUNT	S STUDENT_TYPE
2013	1213	1	476	484	UNDERGRADUATES
2014	1314	1	408	407	UNDERGRADUATES
2015	1415	1	346	355	UNDERGRADUATES
2016	1516	1	332	341	UNDERGRADUATES
2017	1617	1	334	329	UNDERGRADUATES
2018	1718	1	346	362	UNDERGRADUATES
2019	1819	1	272	298	UNDERGRADUATES
2020	1920	1	405	419	UNDERGRADUATES
2021	2021	1	319	281	UNDERGRADUATES
2022	2122	1	325	290	UNDERGRADUATES
2023	2223	1	?	235	UNDERGRADUATES

The table of actual vs predicted values. Showing the predicted value for the number of discontinued students for next fall term.

Tree Ensemble Actual Vs Predicted

Undergraduate Classification Discontinued Spring



First(YEAR_CD)	ACAD_YEAR_CODE	SEMESTER_CD	ACTUAL_COUNT	PREDICTED_COUNT	STUDENT_TYPE
2013	1213	2	1056	849	UNDERGRADUATES
2014	1314	2	1040	806	UNDERGRADUATES
2015	1415	2	1002	724	UNDERGRADUATES
2016	1516	2	972	731	UNDERGRADUATES
2017	1617	2	1007	789	UNDERGRADUATES
2018	1718	2	1131	853	UNDERGRADUATES
2019	1819	2	1333	968	UNDERGRADUATES
2020	1920	2	1115	696	UNDERGRADUATES
2021	2021	2	1302	775	UNDERGRADUATES
2022	2122	2	1622	1013	UNDERGRADUATES
2023	2223	2	?	463	UNDERGRADUATES

The table of actual vs predicted values. Showing the predicted value for the number of discontinued students for next fall term.

Tree Ensemble Actual Vs Predicted

Undergraduate Classification Discontinued Fall



I First(YEAR_CD)	I ACAD_YEAR_CODE	I SEMESTER_CD	I ACTUAL_COUNT	I PREDICTED_COUNT	S STUDENT_TYPE
2013	1314	3	1251	1222	UNDERGRADUATES
2014	1415	3	1276	1245	UNDERGRADUATES
2015	1516	3	1349	1307	UNDERGRADUATES
2016	1617	3	1485	1437	UNDERGRADUATES
2017	1718	3	1506	1457	UNDERGRADUATES
2018	1819	3	1603	1545	UNDERGRADUATES
2019	1920	3	1672	1614	UNDERGRADUATES
2020	2021	3	1935	1830	UNDERGRADUATES
2021	2122	3	1873	1750	UNDERGRADUATES
2022	2223	3	?	1540	UNDERGRADUATES

The table of actual vs predicted values. Showing the predicted value for the number of discontinued students for next fall term.

GRADUATE CLASSIFICATION TO DISCONTINUE

Scorer View

Confusion Matrix

Rows Number : 27118	0 (Predicted)	1 (Predicted)	
0 (Actual)	1794	730	71.08%
1 (Actual)	553	24041	97.75%
	76.44%	97.05%	

Class Statistics

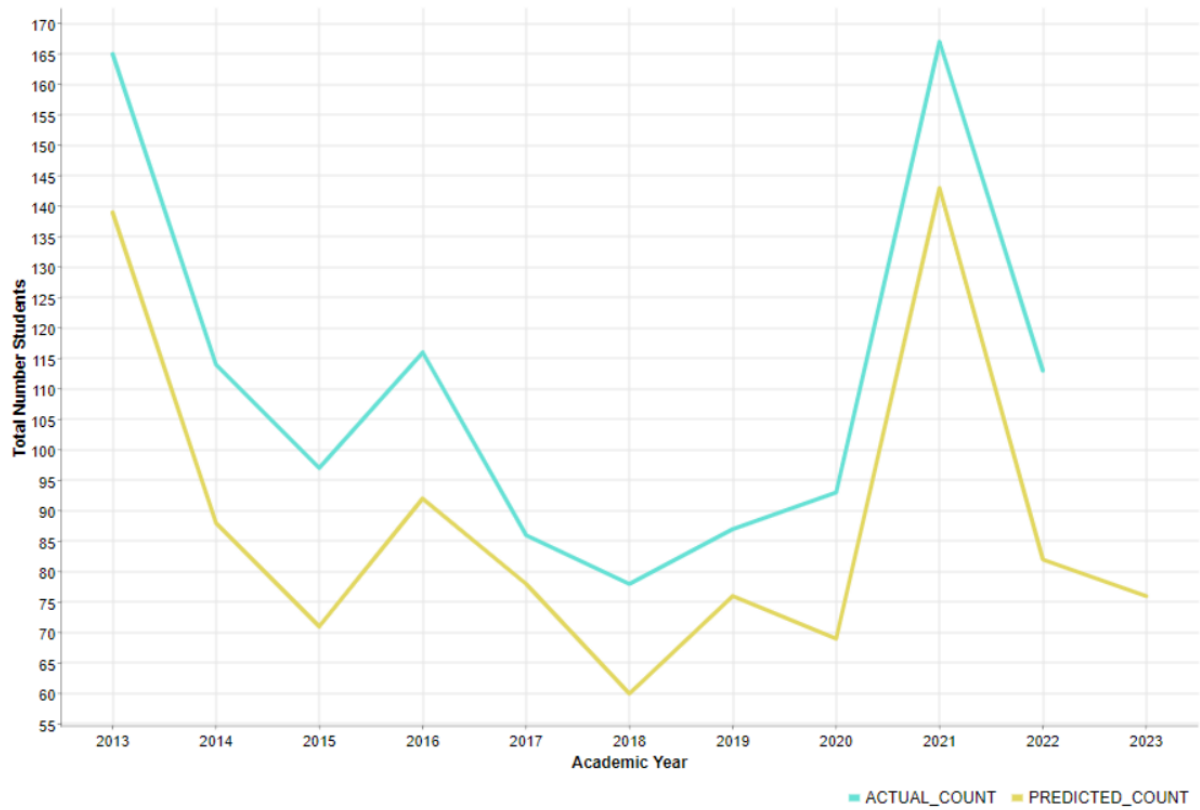
Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
0	1794	553	24041	730	71.08%	76.44%	71.08%	97.75%	73.66%
1	24041	730	1794	553	97.75%	97.05%	97.75%	71.08%	97.40%

Overall Statistics

Overall Accuracy	Overall Error	Cohen's kappa (κ)	Correctly Classified	Incorrectly Classified
95.27%	4.73%	0.711	25835	1283

Tree Ensemble Actual Vs Predicted

Graduate Classification Discontinued Summer



I First(YEAR_CD)	I ACAD_YEAR_CODE	I SEMESTER_CD	I ACTUAL_COUNT	I PREDICTED_COUNT	S STUDENT_TYPE
2013	1213	1	165	139	GRADUATES
2014	1314	1	114	88	GRADUATES
2015	1415	1	97	71	GRADUATES
2016	1516	1	116	92	GRADUATES
2017	1617	1	86	78	GRADUATES
2018	1718	1	78	60	GRADUATES
2019	1819	1	87	76	GRADUATES
2020	1920	1	93	69	GRADUATES
2021	2021	1	167	143	GRADUATES
2022	2122	1	113	82	GRADUATES
2023	2223	1	?	76	GRADUATES

The table of actual vs predicted values. Showing the predicted value for the number of discontinued students for next fall term.

Tree Ensemble Actual Vs Predicted

Graduate Classification Discontinued Spring

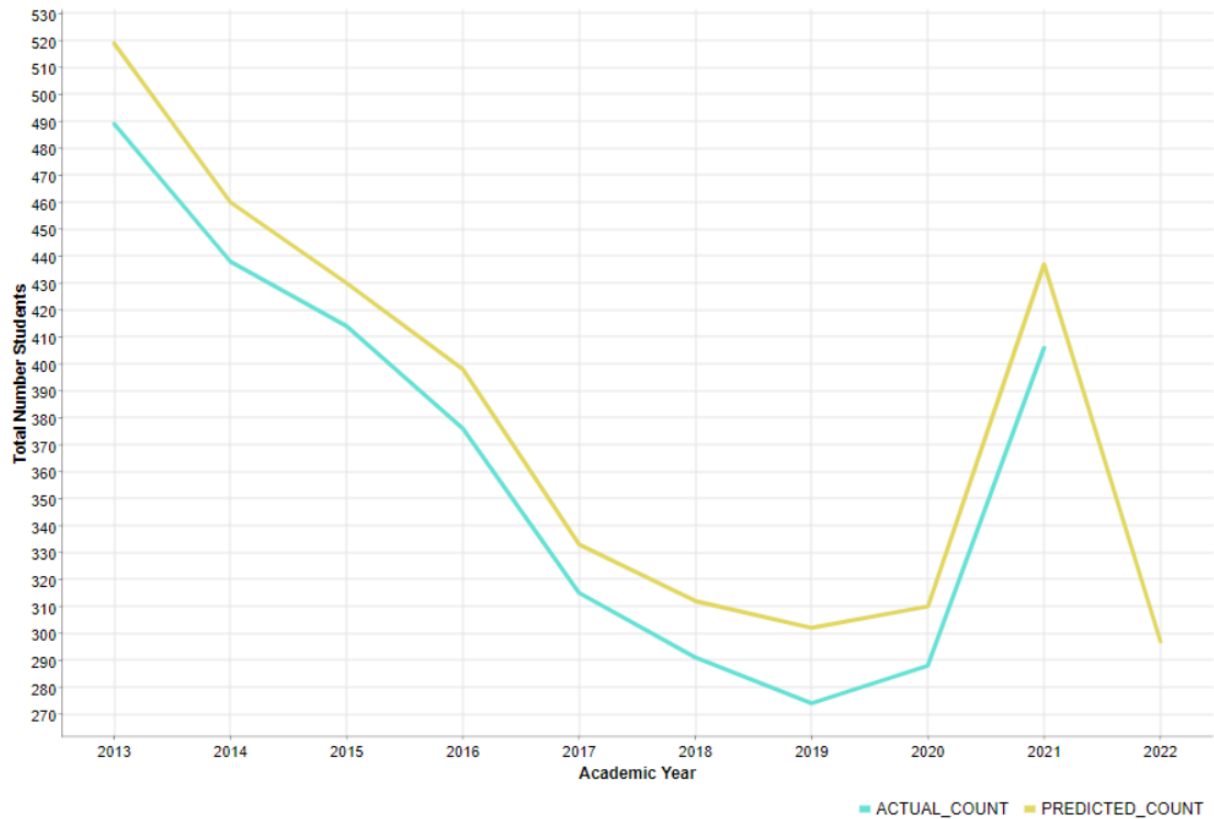


I First(YEAR_CD)	I ACAD_YEAR_CODE	I SEMESTER_CD	I ACTUAL_COUNT	I PREDICTED_COUNT	S STUDENT_TYPE
2013	1213	2	540	521	GRADUATES
2014	1314	2	470	467	GRADUATES
2015	1415	2	459	441	GRADUATES
2016	1516	2	409	394	GRADUATES
2017	1617	2	376	387	GRADUATES
2018	1718	2	360	355	GRADUATES
2019	1819	2	348	330	GRADUATES
2020	1920	2	290	260	GRADUATES
2021	2021	2	394	352	GRADUATES
2022	2122	2	359	276	GRADUATES
2023	2223	2	?	218	GRADUATES

The table of actual vs predicted values. Showing the predicted value for the number of discontinued students for next fall term.

Tree Ensemble Actual Vs Predicted

Graduate Classification Discontinued Fall



I First(YEAR_CD)	I ACAD_YEAR_CODE	I SEMESTER_CD	I ACTUAL_COUNT	I PREDICTED_COUNT	S STUDENT_TYPE
2013	1314	3	489	519	GRADUATES
2014	1415	3	438	460	GRADUATES
2015	1516	3	414	430	GRADUATES
2016	1617	3	376	398	GRADUATES
2017	1718	3	315	333	GRADUATES
2018	1819	3	291	312	GRADUATES
2019	1920	3	274	302	GRADUATES
2020	2021	3	288	310	GRADUATES
2021	2122	3	406	437	GRADUATES
2022	2223	3	?	297	GRADUATES

The table of actual vs predicted values. Showing the predicted value for the number of discontinued students for next fall term.

PROFESSIONAL CLASSIFICATION TO DISCONTINUE

Scorer View

Confusion Matrix

Rows Number : 9431	0 (Predicted)	1 (Predicted)	
0 (Actual)	66	78	45.83%
1 (Actual)	16	9271	99.83%
	80.49%	99.17%	

Class Statistics

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
0	66	16	9271	78	45.83%	80.49%	45.83%	99.83%	58.41%
1	9271	78	66	16	99.83%	99.17%	99.83%	45.83%	99.50%

Overall Statistics

Overall Accuracy	Overall Error	Cohen's kappa (κ)	Correctly Classified	Incorrectly Classified
99.00%	1.00%	0.579	9337	94

Tree Ensemble Actual Vs Predicted

Professional Classification Discontinued Summer



I First(YEAR_CD)	I ACAD_YEAR_CODE	I SEMESTER_CD	I ACTUAL_COUNT	I PREDICTED_COUNT	S STUDENT...
2013	1213	1	3	1	PROFESSIONAL
2014	1314	1	4	4	PROFESSIONAL
2015	1415	1	10	7	PROFESSIONAL
2016	1516	1	7	4	PROFESSIONAL
2017	1617	1	8	7	PROFESSIONAL
2018	1718	1	10	6	PROFESSIONAL
2019	1819	1	6	5	PROFESSIONAL
2020	1920	1	9	6	PROFESSIONAL
2021	2021	1	6	5	PROFESSIONAL
2022	2122	1	15	11	PROFESSIONAL
2023	2223	1	?	1	PROFESSIONAL

Tree Ensemble Actual Vs Predicted

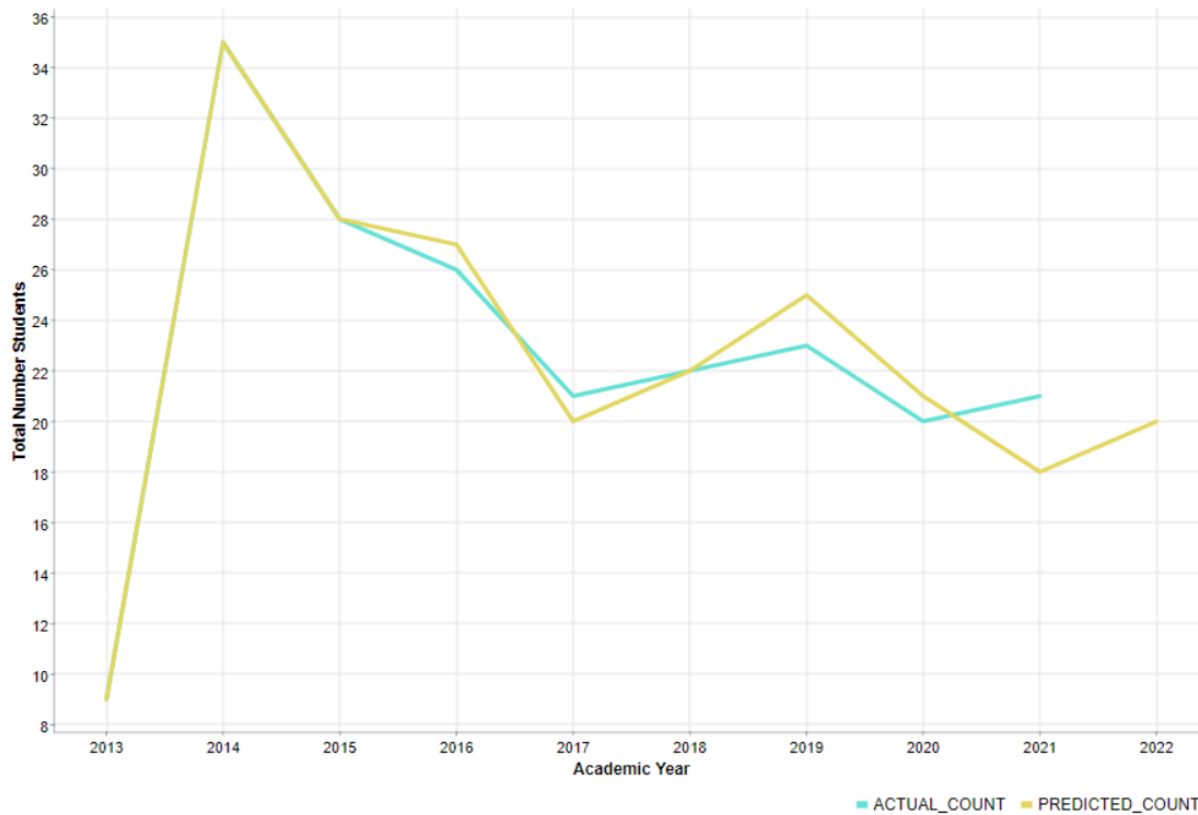
Professional Classification Discontinued Spring



I First(YEAR_CD)	I ACAD_...	I SEMES...	I ACTUAL_COUNT	I PREDICTED_COUNT	S STUDENT...
2013	1213	2	12	7	PROFESSIONAL
2014	1314	2	7	3	PROFESSIONAL
2015	1415	2	27	28	PROFESSIONAL
2016	1516	2	18	20	PROFESSIONAL
2017	1617	2	21	20	PROFESSIONAL
2018	1718	2	22	16	PROFESSIONAL
2019	1819	2	27	15	PROFESSIONAL
2020	1920	2	10	8	PROFESSIONAL
2021	2021	2	17	12	PROFESSIONAL
2022	2122	2	36	23	PROFESSIONAL
2023	2223	2	?	5	PROFESSIONAL

Tree Ensemble Actual Vs Predicted

Professional Classification Discontinued Fall



I First(YEAR_CD)	I ACAD_YEAR_CODE	I SEMESTER_CD	I ACTUAL_COUNT	I PREDICTED_COUNT	S STUDENT...
2013	1314	3	9	9	PROFESSIONAL
2014	1415	3	35	35	PROFESSIONAL
2015	1516	3	28	28	PROFESSIONAL
2016	1617	3	26	27	PROFESSIONAL
2017	1718	3	21	20	PROFESSIONAL
2018	1819	3	22	22	PROFESSIONAL
2019	1920	3	23	25	PROFESSIONAL
2020	2021	3	20	21	PROFESSIONAL
2021	2122	3	21	18	PROFESSIONAL
2022	2223	3	?	20	PROFESSIONAL

LAW CLASSIFICATION TO DISCONTINUE

Scorer View



Confusion Matrix

Rows Number : 625	0 (Predicted)	1 (Predicted)	
0 (Actual)	14	21	40.00%
1 (Actual)	1	589	99.83%
	93.33%	96.56%	

Class Statistics

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
0	14	1	589	21	40.00%	93.33%	40.00%	99.83%	56.00%
1	589	21	14	1	99.83%	96.56%	99.83%	40.00%	98.17%

Overall Statistics

Overall Accuracy	Overall Error	Cohen's kappa (κ)	Correctly Classified	Incorrectly Classified
96.48%	3.52%	0.545	603	22

Tree Ensemble Actual Vs Predicted



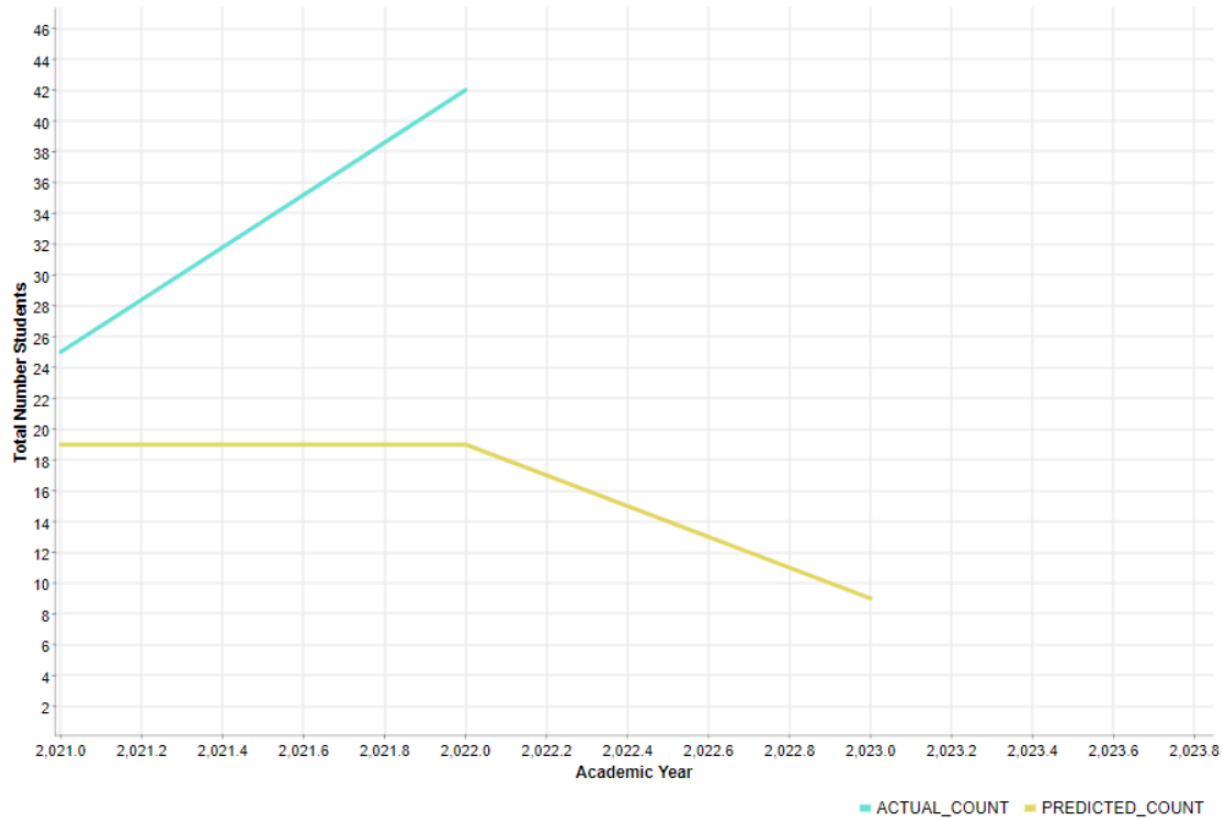
Law Classification Discontinued Summer



First(YEAR_CD)	ACAD_YEAR_CODE	SEMESTER_CD	ACTUAL_COUNT	PREDICTED_COUNT	STUDE...
2021	2021	1	10	6	LAW
2022	2122	1	5	2	LAW

Tree Ensemble Actual Vs Predicted

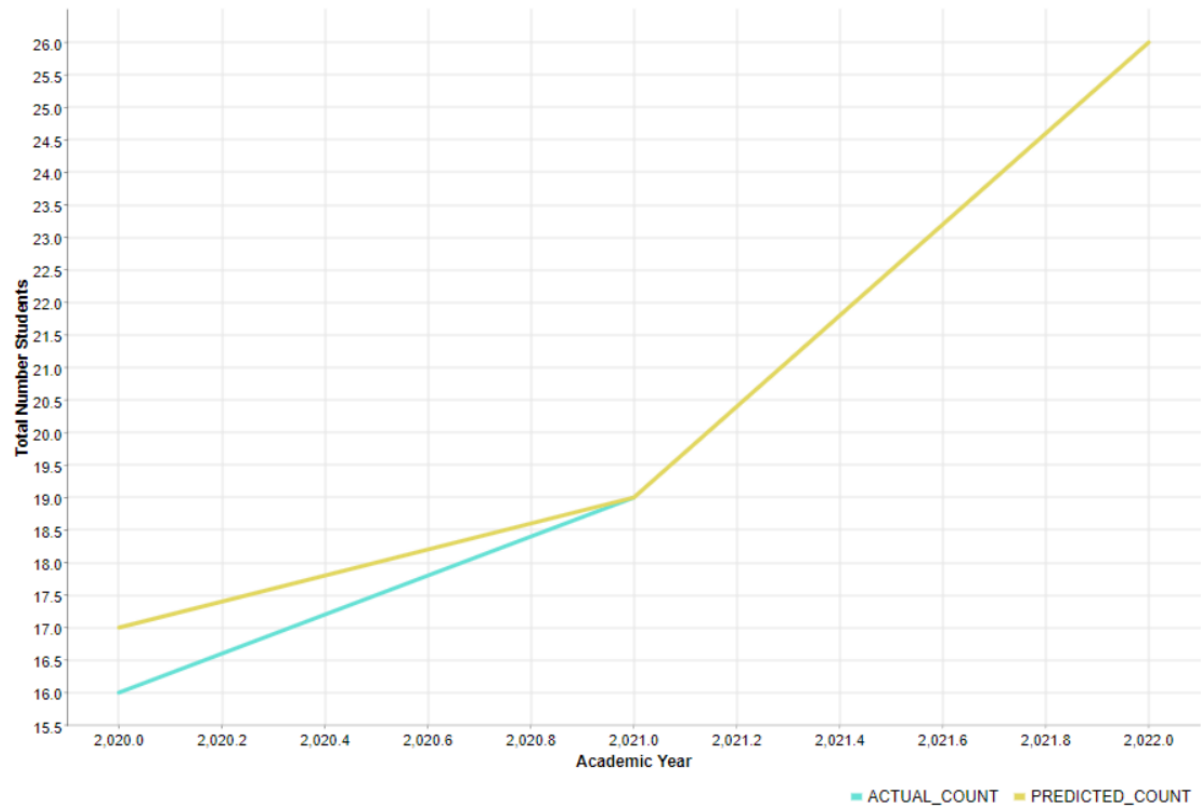
Law Classification Discontinued Spring



First(YEAR_CD)	ACAD_YEAR_CODE	SEMESTER_CD	ACTUAL_COUNT	PREDICTED_COUNT	STUDE...
2021	2021	2	25	19	LAW
2022	2122	2	42	19	LAW
2023	2223	2	?	9	LAW

Tree Ensemble Actual Vs Predicted

Law Classification Discontinued Fall



I	First(YEAR_CD)	I	ACAD_YEAR_CODE	I	SEMESTER_CD	I	ACTUAL_COUNT	I	PREDICTED_COUNT	S	STUDENT_TYPE
	2020		2021		3		16		17		LAW
	2021		2122		3		19		19		LAW
	2022		2223		3		?		26		LAW

FINAL RESULTS

I ACAD_YEAR_CODE	I SEMESTER_CD	S STUDENT_TYPE	I Sum(PREDICTED_COUNT)
2223	1	GRADUATES	76
2223	1	PROFESSIONAL	1
2223	1	UNDERGRADUATES	235
2223	2	GRADUATES	218
2223	2	LAW	9
2223	2	PROFESSIONAL	5
2223	2	UNDERGRADUATES	463
2223	3	GRADUATES	297
2223	3	LAW	26
2223	3	PROFESSIONAL	20
2223	3	UNDERGRADUATES	1540

This is the total classification for the different student levels based on semesters.

This is the total predicted number of student to discontinue in the academic year 2223 by student level description

Row ID	L ACAD_YR_CD	D TOTAL_STUDENTS	S TYPE
?_Row10	2223	-2,890	DISCONTINUED STUDENTS
Row10	2223	-9,758	GRADUATED STUDENTS
Row10_dup	2223	36,688	TOTAL STUDENTS

The total number of students can be determined by counting the registration and enrollment figures, while identifying graduate students can be done by checking if the STUDENT_CURR_DEG_IND is "Y," indicating the awarding of a degree. Subtracting these graduate students from the overall enrollment count for the academic year of 2223, along with the predicted number of discontinued students, allows us to calculate the expected total number of students continuing into the next fall term, given the known values.

Row ID	D CONTINUING_STUDENTS
Row0	24,040

Total number of students to continue into the next fall term.

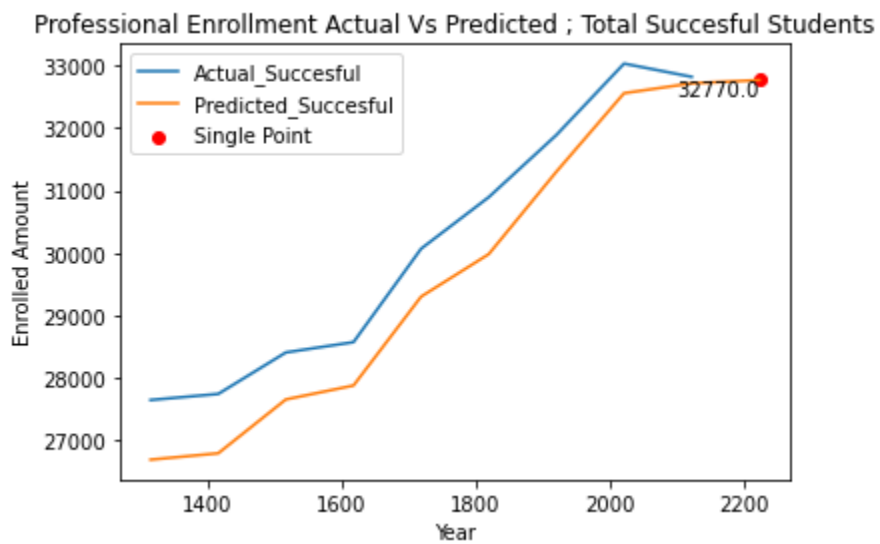
PYTHON RANDOM FOREST REGRESSION

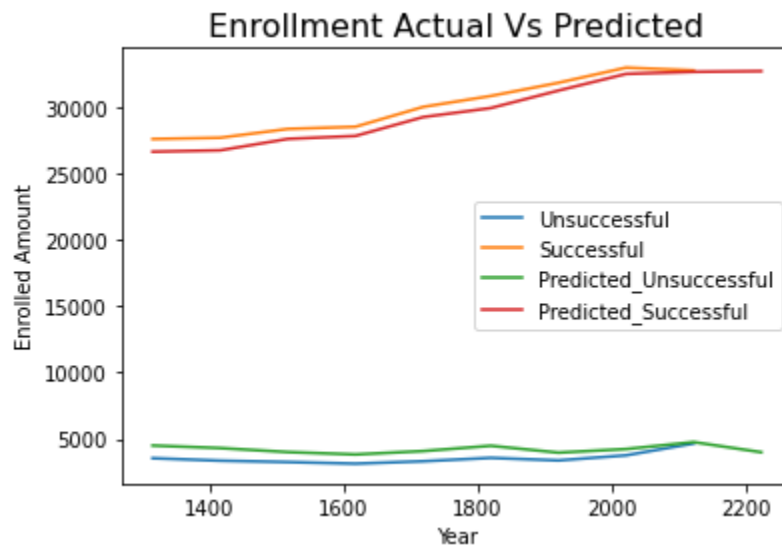
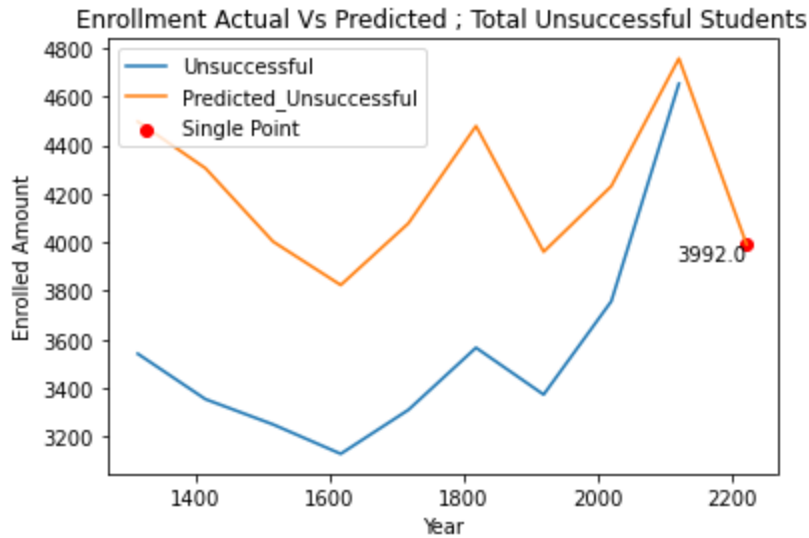
In this python script we have used a random forest regression to make the classification. Random Forest Regression is a supervised machine learning algorithm that combines multiple decision trees trained on randomly sampled subsets of data and features to predict continuous numerical values. By aggregating the predictions from individual trees, it provides robust and accurate predictions while mitigating overfitting and handling missing values. Here we have tested the data using the "SUCCESSFUL_OUTCOME_TARGET_BY_YEAR" and have made an overall classification of all students for the academic year of 2223.

Mean Squared Error: 0.05348813218317835

R-squared: 0.6836633727539814

	Year	Successful	Unsuccessful	Predicted_Successful	Predicted_Unsuccessful
0	2223	NaN	NaN	32770.0	3992.0
1	2122	32822.0	4653.0	32718.0	4757.0
2	2021	33034.0	3757.0	32560.0	4231.0
3	1920	31890.0	3373.0	31302.0	3961.0
4	1819	30894.0	3566.0	29981.0	4479.0
5	1718	30068.0	3310.0	29300.0	4078.0
6	1617	28570.0	3129.0	27875.0	3824.0
7	1516	28405.0	3250.0	27652.0	4003.0
8	1415	27740.0	3354.0	26789.0	4305.0
9	1314	27643.0	3541.0	26687.0	4497.0





Final Notes & Discussion

When reviewing the performance metric of our compared model we see that while we have high overall accuracy of the classification, the accuracy for discontinuation is poor. This could be an indication that there is an imbalance dataset or a specific issue with that particular class. Here are a few possible explanations of what the cause of the issues may be and a few suggestions on next steps to remedy this problem.

CAUSE

Imbalanced dataset: The class with poor accuracy may have significantly fewer instances compared to the other classes. As a result, the model may prioritize accuracy on the majority

class(es) while struggling to perform well on the minority class. In such cases, overall accuracy can be misleading, and it's important to consider other evaluation metrics like precision, recall, or F1 score to assess performance more comprehensively.

SUGGESTION

Balance the data by Implementing a method of oversampling the minority class. Possible methods such as Random Over-sampling, Random Under-sampling, Cluster-based Over-sampling, Borderline-SMOTE, ADASYN, SMOTE-ENN, and SMOTE-Tomek are alternative techniques to consider.

CAUSE

Class-specific challenges: The class with poor accuracy may have inherent complexities or characteristics that make it harder to classify correctly. It could be due to similarities with other classes, insufficient training examples, or insufficient representation of important features for that class. In these cases, it may be necessary to focus on improving the model's performance specifically for that class through techniques such as collecting more data, data augmentation, or adjusting the training strategy.

SUGGESTION

Improve the feature selection process. The most widely used methods include feature importance, recursive feature elimination (RFE), permutation importance, drop-column importance, SelectFromModel, and LightGBM's feature importance.

CAUSE

Biased training: The model might have been trained on a biased dataset, leading to a discrepancy in performance across different classes. If the training data is not representative of the real-world distribution of classes, the model may struggle with accurate classification for underrepresented classes. Addressing this issue requires careful data collection and preprocessing to ensure balanced and diverse training samples. As we have seen in the previous analysis the classification threshold was left-skewed and had a non-normal probability distribution and even with adjustment ratio based threshold tuning may still lead to uneven classifications and under/over-prediction issues.

SUGGESTION

In summary, a high overall accuracy with a poor accuracy for one class suggests that the model is performing well on most classes but encountering difficulties with a specific class. We can try to adjust the sensitivity of thresholds to include more or less of a certain class.

A possible suggestion is to adjust Therefore, it is advisable to explore alternative approaches, such as utilizing the Probability Density Function (PDF), to accurately represent the distribution. By integrating the PDF over a specified range of values, the proportion of samples falling within that range can be determined, allowing for more precise threshold adjustment and improving classification performance.

CAUSE

To ensure a comprehensive analysis of student data for the academic year 2223, including summer student data and obtaining accurate calculations of the total number of students set to

discontinue or continue, it is crucial to load the data after the census snapshot date using `reg_snapshot_key = "Census"`. Unfortunately, it was only discovered at the end of our analysis that approximately 4000 students were missing from our data, resulting in a sudden drop in enrollment numbers and highlighting the significance of loading complete data for reliable insights.

SUGGESTION

Generate predictions using the data after the summer census date in order to get accurate predictions. If this is not possible and projection you may need to train a model removing all students from summer and make classifications only on spring and fall enrolled students.